

Module 5: Project Estimation, COCOMO & CMMI

Function Points, Basic/Intermediate COCOMO, CMMI 5–Levels, SCM

Module V: Project Estimation, COCOMO & CMMI — Topics Covered

1. Function Point Analysis (Albrecht UFP & VAF)
2. COCOMO I (Organic, Semidetached, Embedded Modes)
3. SEI CMMI 5–Level Maturity Framework
4. Software Configuration Management (SCM Baselines)
5. Software Maintenance: 4 Types & Lehman's Laws

1. Basic COCOMO Estimation Formulas (Barry Boehm)

$$\text{Effort (Person-Months)} = a_b \times (\text{KLOC})^{b_b}$$

$$\text{Development Time (Months)} = c_b \times (\text{Effort})^{d_b}$$

Project Mode	Team & Project Description	a_b	b_b	c_b	d_b
Organic	Small experienced team, well-understood domain (< 50 KLOC).	2.4	1.05	2.5	0.38
Semidetached	Medium team, mixed experience, medium size (50 – 300 KLOC).	3.0	1.12	2.5	0.35
Embedded	Tight hardware/software constraints, mission-critical.	3.6	1.20	2.5	0.32